

PROJECT DESIGN PHASE

SOLUTION ARCHITECTURE

Date	28.10.2025
Team id	NM2025TMIDo4506
project name	Laptop Request Catalog Item
Maximum marks	4 marks

Solution Architecture:

Goals of the Architecture:

1. Automate the laptop request process using the Service Catalog.
2. Ensure approval and fulfillment workflows align with IT asset policies.
3. Maintain accurate asset tracking through CMDB and Asset tables.
4. Reduce manual effort in request handling and status updates.

Key Components

1. Service Catalog Item (Laptop Request) – user-facing catalog form.
2. Workflow / Flow Designer – automates approval, procurement, and delivery.
3. Tables: .sc_req_item – stores each laptop request.

- .alm_asset – tracks laptops in stock.
- .cmdb_ci_computer – maintains configuration items (CIs).
- 4. Business Rules & Script Includes – auto-assign requests and validate eligibility.
- 5. Approvals – routed to managers or department heads.
- 6. Notifications – for status updates and fulfillment alerts.

Development Phases:

1. Design the Laptop Request catalog form (fields: Laptop Type, Purpose, Location, Needed By Date).
2. Configure the Flow Designer for approval and fulfillment workflow.
3. Integrate with Asset Management to check laptop availability.
4. Automate Procurement Requests if stock is insufficient.
5. Test various scenarios: approved, rejected, fulfilled, and out-of-stock.

Solution Architecture Description:

The Laptop Request Catalog Item solution architecture is designed to simplify and automate the process of requesting and assigning laptops within an organization using the ServiceNow platform.

Employees can request laptops through the Service Catalog, triggering a workflow that handles approval, inventory validation, and fulfillment.

The workflow checks laptop availability in the Asset table (alm_asset), routes the request for managerial approval, and either assigns an available laptop or raises a procurement request if stock is insufficient. Business Rules ensure data consistency and automatic status updates throughout the process.

This architecture enhances efficiency, reduces manual workload, and ensures accurate asset tracking through integration with CMDB and Procurement systems, promoting transparency and accountability in IT service operations.

EXAMPLE - SOLUTION ARCHITECTURE DIAGRAM.

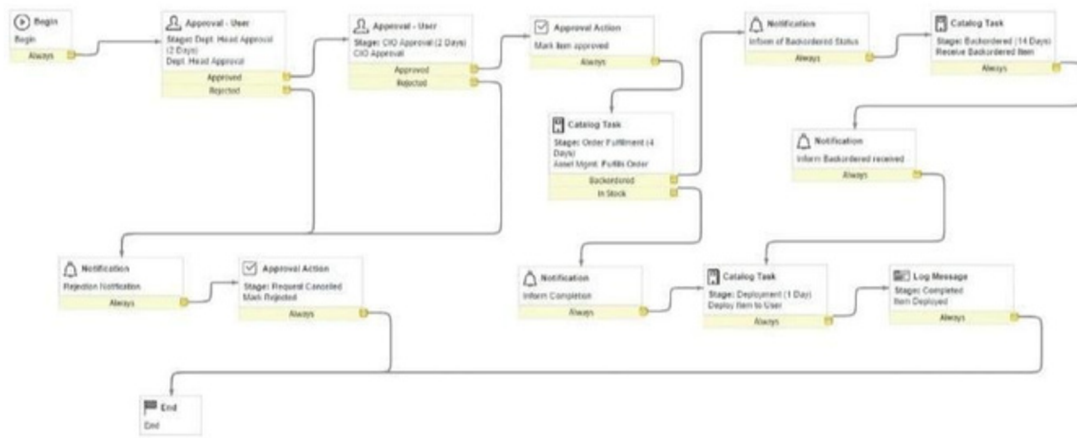


FIG 1: Architecture and workflow of the Laptop Request Catalog Item solution.